Rooms - class diagram

Each dotted region is a package. Getters, setters and most private helpers are left out.

Arrows	
solid, hollow triangle	extends
dashed, hollow triangle	implements
filled diamond	owns (composition)
hollow diamond	holds a reference
dashed, open arrow	uses / creates

init + Main (default package)

Main
+main(args)

Game
+run()
-newPlayer(): Player
-play(player)

Menu
+promptChoice(): Action

inner

«enum» **Menu.Action**
NEW_GAME, CONTINUE, EXIT

player

PlayerController
-player: Player
+printHelp()
+playTurn(): boolean
-handle(input): boolean
-execute(command)
-undo()

Player *Serializable*
-stats: Map<String, Integer>
-discoveredRoomIds: Set<Integer>
-currentRoom, previousRoom: Room
+getStatValue(stat): int
+pickUpItem(item) / discardItem(item): boolean
+undoPickUp(...) / undoDiscard(...)
+move(direction): boolean
+retreat() / returnTo(room)
+showMap()
+record(command) / takeLastCommand()
+forgetHistory()

PlayerView
+showStats(player)
+showInventory(player)
+showItemsInRoom(room)

creates

utilities map and saves

MapLoader
builds every Room from rooms.csv
+loadMap(filePath): Map<Integer, Room>
-parseRoom(line): Room
-validateLinks(rooms)
-validateEveryRoomCanBeEntered(rooms)

SaveManager
-saveFile: Path
+useFile(file)
+saveExists(): boolean
+save(player): boolean
+load(): Player
+deleteSave()

MapView
+print(rooms, discovered, currentId)

history *

roomMap *

inventory *

commands

«interface» **Command**
+execute(): boolean
+undo()

MoveCommand
-player: Player
-direction: Direction
-from: Room

PickUpItemCommand
-player: Player
-item: Item
-statsBefore: Map
-roomIndex: int

DiscardItemCommand
-player: Player
-item: Item
-statsBefore: Map
-inventoryIndex: int

enemy

Combat
+confront(player)
+damage(strength, weapons): int
-fight(player, enemy, health): int

fights

enemies *

Enemy *Serializable*
-name: String
-maxHealth, health: int
-strength, agility: int
+heal()
+strikesFirstAgainst(agility): boolean
+damageBeforeFalling(damage, agility): long

room

«enum» **Direction**
NORTH, EAST, SOUTH, WEST
-columnStep, rowStep: int

Room *Serializable*
-id: int
-name, description: String
-combatMessage, winMessage: String
-north, east, south, west: int
-isWinRoom, isSecret: boolean
+unmetCondition(player): Optional
+getExit(direction): int

conditions *

RoomCondition *Serializable*
-stat: String
-minValue: int
-message: String

items *

items

Item *Serializable*
-name: String
-stats: Map<String, Integer>

Weapon
-damage: int

utilities used by most classes, arrows left out

Console
-scanner: Scanner
+useInput(input)
+hasInput(): boolean
+readLine(): String
+confirm(question): boolean

Logic
+printToUser(text)

Config
file names, stat names, tunable numbers
«enum» Column: map columns in order

myExceptions

InputEndedException
extends RuntimeException
thrown by Console.readLine()

ParsingLevelException
extends Exception
thrown by MapLoader.loadMap()